Analytical Chemistry Cumulative Exam
October 19, 2012

Some equations.

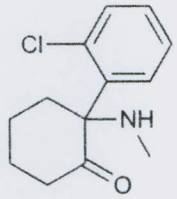
$$z = \frac{x - \mu}{\sigma}$$

$$\mu = \bar{x} \pm \frac{ts}{\sqrt{n}}$$

$$\mu = \frac{1}{2}\sum c_i z_i^2$$

1. What is the approximate pH of a 1.0×10^{-8} M solution of H_2SO_4?

2. Buffer solutions with constant ionic strength are necessary for many types of biochemical separations. Explain how you would make a 1.0 L sodium acetate buffer, pH 5.0, with an ionic strength of 0.1 M. The pK_a for acetic acid is 4.75.

3. Ketamine (shown below) is a drug used for general anesthesia. It is also commonly sold on the street as "Special K." It has been found in wastewater from different municipal wastewater treatment plants. Ketamine has a pK_a of 7.5 and a log P of ~ 3.0 (2.88 actually, but use 3.0). Using this information, describe how you would extract this compound from wastewater such that you could perform a chromatographic analysis on the extracted sample. Be sure to address issues such as recovery in your answer. You can use either gas or liquid chromatography to analyze your extracted sample. GC and LC methods have been described for this compound.

If you don't know what log P is, then make an educated guess and justify your answer.